Biostat 625 - Computing with Big Data

R package

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Why R package

- Code organization - modular design
- Software distribution
- Bundling code with documents
- Version control

Create package

```
dat = rnorm(100)
square = function(x) {
  return (x^2)
}
package.skeleton("mypackage",list =
c("dat","square"))
```

R package folder structure

- NAMESPARE – import / export
- DESCRIPTION
- R/ - R code
- man/ - documents

- src/ - C++ source code
- data/ - sample data
- vignettes/ - tutorial
- …

Deployment

- At terminal command line
  - `R CMD build`
  - `R CMD INSTALL`
  - `R CMD check`

With `devtools` package

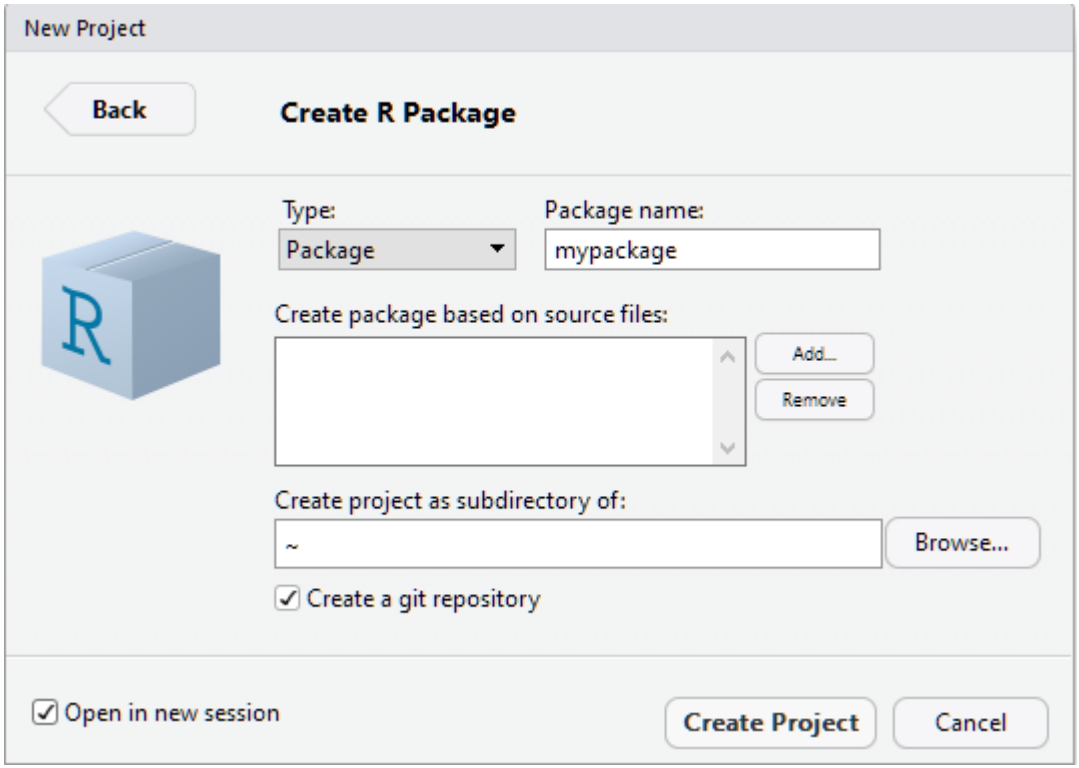
- Create package

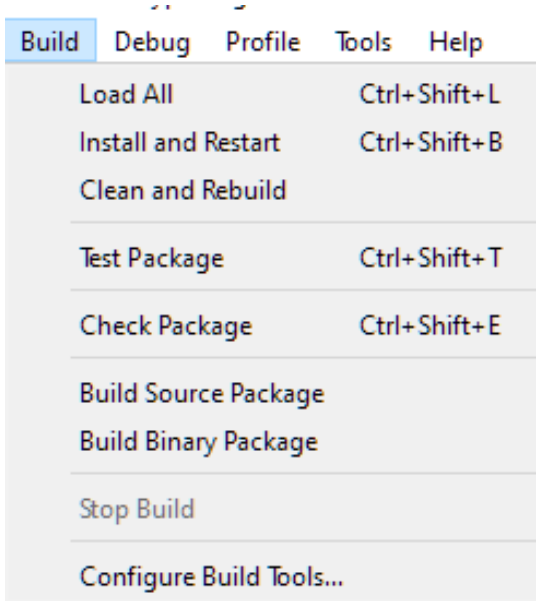
```
devtools::create("mypackage")
```

- Deployment
  - `devtools::build()`
  - `devtools::install()`
  - `devtools::check()`

With RStudio

- File -> New Project





Publish R package

- Publish on personal website – informal
- Publish on CRAN – formal review process
- Publish on Bioconductor – even more rigorous
- Publish on GitHub
  - No review needed
  - Convenient installation
  - Version control, collaboration, issue tracking, etc
  - Integration with RStudio

R package on GitHub

- Create a new repository on GitHub.
- Create a new R package project in RStudio with a local git repository.
- Commit and push changes to GitHub

```
git add *

git commit

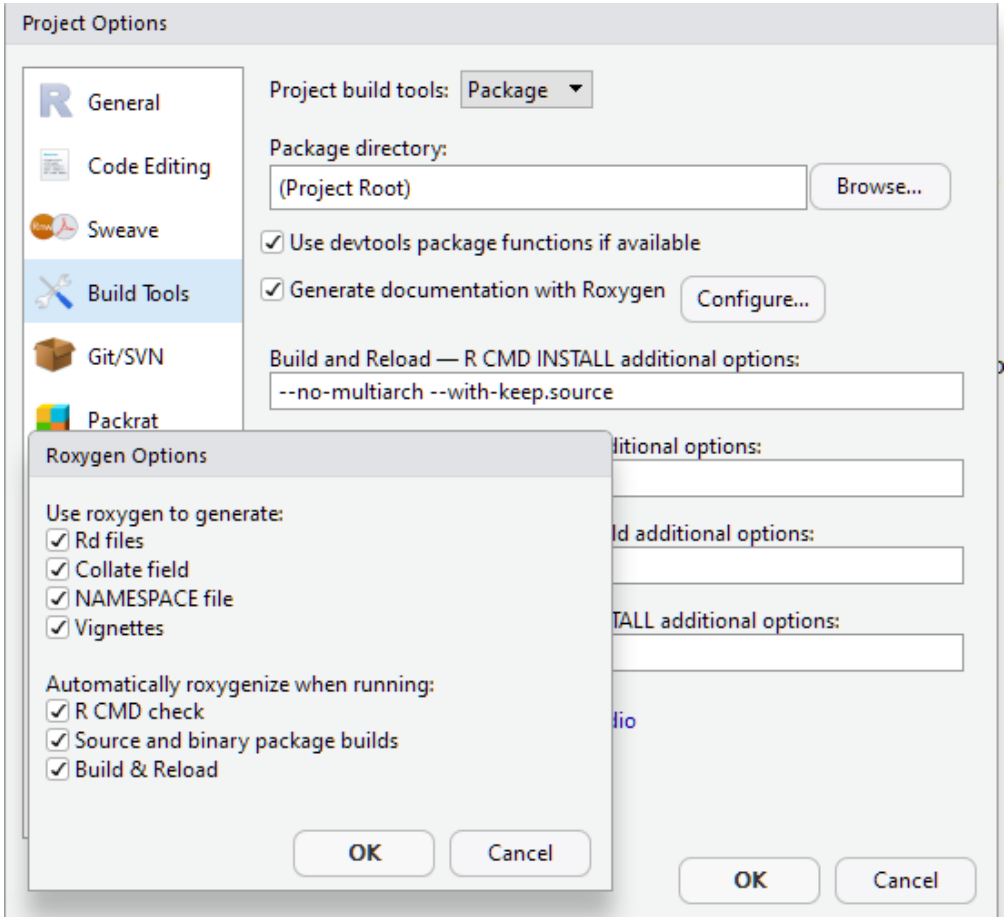
git remote add origin
git@github.com:YourGitHubAccount/YourGitHubRepository

git push -u origin master
```

- Install package from GitHub

```
devtools::install_github("YourGitHubAccount/YourGitHubR
epository")
```

Documentation using ROxygen2



Documentation using ROxygen2 (cont.)

```
#'Square
#'
#'Gets the square of a number
#'
#'@param x input value
#'
#'@return the square of x
#'
#'@examples
#'square(3)
#'
#'@export
#'
square = function(x) {
    return (x^2)
}
```

Help page

square.Rd

Preview on Save | Preview

```
% Generated by roxygen2: do not edit by hand
% Please edit documentation in R/square.R
\name{square}
\alias{square}
\title{Square}
\usage{
square(x)
}
\arguments{
\item{x}{input value}
}
\value{
the square of x
}
\description{
Gets the square of a number
}
\examples{
square(3)

}
```

R: Square | Find in Topic

square {test.package} R Documentation

Square

Description

Gets the square of a number

Usage

```
square(x)
```

Arguments

x	input value

Value

the square of x

Examples

```
square(3)
```

[Package *test.package* version 0.0.0.9000 Index]

DESCRIPTION

DESCRIPTION

```
Package: test.package
Title: Test R package
Version: 0.0.0.9000
Maintainer: Hui Jiang <jianghui@umich.edu>
Description: Demonstrate R package compilation and installation.
License: GPL-2
Encoding: UTF-8
LazyData: true
RoxygenNote: 6.1.1
```

NAMESPARE

NAMESPACE

```
# Generated by roxygen2: do not edit by hand

export(square)
```

Vignettes

- Tutorials on how to use your package.
- Can be written in Latex (compile with **`Sweave`** or **`knitr`**) or R Markdown (compile with **`knitr`**).
- R Markdown (https://rmarkdown.rstudio.com) is an extension of Markdown, which allows R code (and output) be included in a document, which also facilitates reproducible research report.

Steps

- Create a **`vignettes`** subfolder.
- Add a **`.Rmd`** file in that folder.
- Add the following lines to the **`DESCRIPTION`** file:

```
Suggests: knitr, rmarkdown
VignetteBuilder: knitr
```

- Do all the above in one step

```
usethis::use_vignette()
```

- To build package

```
devtools::build_vignettes()
```

- To install package with vignettes

```
devtools::install(build_vignettes = T)
devtools::install_github("YourGitHubAccount/YourGitHubRepos
itory", build_vignettes = T)
```

- To browse vignettes

```
browseVignettes("YourPackage")
```

Vignettes example

tutorial.Rmd

Knit

```
---
title: "Tutorial"
output: html_vignette
vignette: >
  %\VignetteIndexEntry{test.package tutorial}
  %\VignetteEngine{knitr::rmarkdown}
  \usepackage[utf8]{inputenc}
---

```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
library(test.package)
```

To use the function `square`:

```{r}
square(1:3)
```

To include plots:

```{r, echo=FALSE}
plot(1:10, square(1:10))
```
```

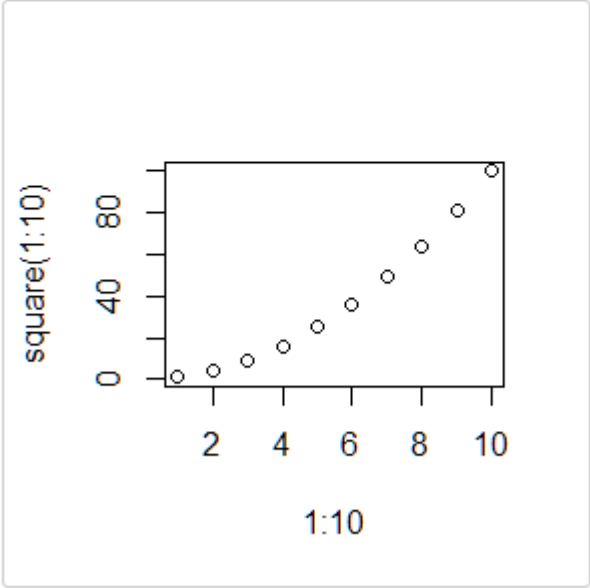
Tutorial

To use the function `square`:

```
square(1:3)
```

```
## [1] 1 4 9
```

To include plots:



Unit testing

- Set up unit testing

```
usethis::use_testthat()
```

- Set up test case

```
usethis::use_test("square")
```

- Edit test case

test-square.R

Source on Save

```
test_that("square works", {
  expect_equal(square(2), 4)
  expect_equal(square(1:3), c(1,4,9))
})
```

- Run unit tests

```
> devtools::test()
Loading test.package
Testing test.package
√ |  OK F W S | Context
√ |   2       | square [0.1 s]

== Results
=================================================================
Duration: 0.1 s

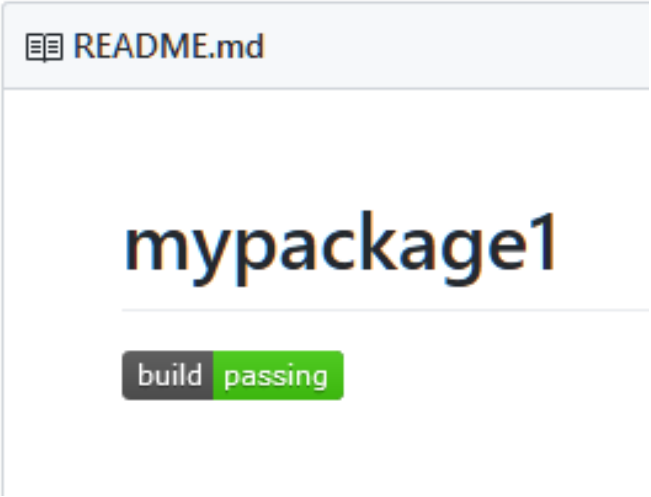
OK:       2
Failed:   0
Warnings: 0
Skipped:  0
```

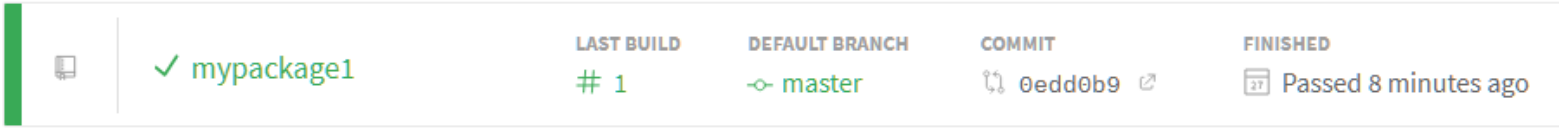
Continuous integration

- Set up continuous integration

  `usethis::use_travis()`

- Add Travis badge to `README.md`
- Every commit will be automatically tested.



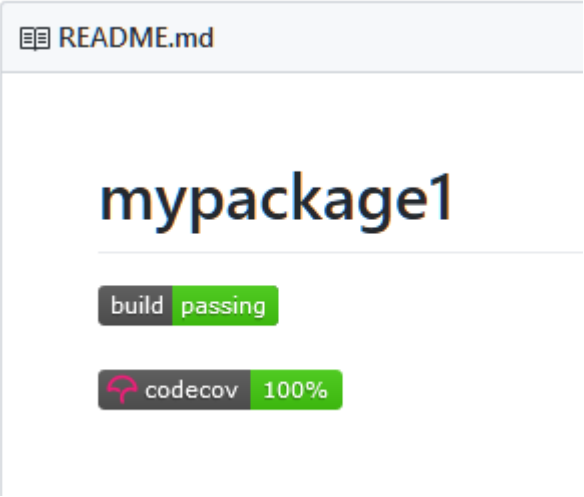


Code coverage

- Set up code coverage

  `usethis::use_coverage()`

- Add codecov badge to `README.md`
- Add coverage test to `.travis.yml`
- Commit and push changes.



Files	≡	●	●	●	Coverage
R/hello.R	1	1	0	0	100.00%
Project Totals (1 files)	1	1	0	0	100.00%

R package with Rcpp

- Set up Rcpp

```
usethis::use_rcpp()
```

- Edit files as suggested.
- Edit cpp files.

```
Rcpp::compileAttributes()
pkgbuild::compile_dll()
devtools::document()
```

Readings

- Wickham H. (2015). R packages. O'Reilly. http://r-pkgs.had.co.nz/
- https://kbroman.org/pkg_primer/
- https://cran.r-project.org/doc/contrib/Leisch-CreatingPackages.pdf
- https://support.rstudio.com/hc/en-us/articles/200486488?version=1.1.463&mode=desktop
- https://cran.r-project.org/doc/contrib/Leisch-CreatingPackages.pdf
- https://cran.r-project.org/doc/manuals/R-exts.html
- http://kbroman.org/pkg_primer/pages/docs.html
- https://rmarkdown.rstudio.com/index.html
- https://guides.github.com/features/mastering-markdown/
- https://help.github.com/en/github/writing-on-github/basic-writing-and-formatting-syntax#quoting-code